

Infosys Springboard Virtual Internship 7.0 Completion Report

Batch Number:	Batch 7.0 / Group 1	Start Date:	July 2026
Project Domain:	Artificial Intelligence & Maritime Logistics	Internship Duration:	3 Months (12 Weeks)
Primary Tech Stack:	React 19, Django REST Framework, Python 3.14, Scikit-Learn, Pytest	Live App Status:	Production Verified & Cloud Deployed

Project Team Member Roster (6 Members):

#	Team Member Name	Assigned Project Role	Core Technical Deliverables
1	Naresh Ramavath	Team Lead & System Architect & RAG	System Architecture, Django REST backend framework, API contracts, Hybrid RAG pipeline, M1–M4 integration
2	Nutan Jadhav	Frontend Lead Engineer	React client application, SPA routing, multi-role dashboard portals (Shipper, Agent, Customs, Admin)
3	Biswajit Pradhan	Machine Learning Engineer	5,000-shipment historical training dataset, Gradient Boosted regression model, price benchmark analysis
4	Syona Jain	Frontend Lead Engineer	Interactive UI/UX design, responsive quote & booking flows, Leaflet routing maps, Chart.js analytics
5	Arpita Dash	Backend & Telemetry Developer	Open-Meteo marine weather agent, sea route waypoints, 6-hr coordinate cache, 5-Factor MCDA risk engine
6	Ishanvi Chavan	QA & DevOps Specialist	Automated Pytest test suite, carrier multi-tenant isolation validation, Vercel & Cloudflare deployment, CI/CD

1. Project Title

Agentic AI for Maritime Freight Pricing and Route Optimization

2. Project Objective

International ocean freight quoting remains heavily bottlenecked by manual workflows: freight forwarders exchange unstructured emails, check disparate static tariff sheets, manually verify customs document checklists, and cross-reference port congestion reports, resulting in standard quote turnarounds of 2 to 4 business days. The objective of this 3-month internship project was to engineer **FreightAI**—an end-to-end intelligent quotation and risk platform that automates pricing, maritime weather delay telemetry, statutory customs compliance, and carrier booking verification into a sub-3-second transaction pipeline. By integrating deterministic rule engines, trained Gradient Boosted price regressors, and a Hybrid BM25+Vector Retrieval-Augmented Generation (RAG) customs service, FreightAI eliminates pricing opacity, identifies route risks prior to booking, and enforces governed carrier verification workflows.

3. Project Description in Detail

FreightAI was implemented across 4 structured engineering milestones (M1 to M4) delivering a complete multi-role logistics platform:

- Milestone 1 (Foundational Routing & Rule-Based Pricing):** Built core freight domain entities (Shipment, Route, Cargo, BaseRate). Engineered an automated chargeable weight resolver calculating volumetric weight against actual gross weight (using standard maritime ratios: 1 CBM = 167 kg for air and full container load equivalents). Integrated port-to-port nautical distance matrices, transit ETA estimation, and deterministic base ocean rates with bunker adjustment factors (BAF) and peak season surcharges.
- Milestone 2 (AI/ML Freight Price Prediction):** Assembled and cleaned a historical dataset of 5,000 real maritime shipments across major trade corridors (Transpacific, Asia-Europe, Intra-Asia, Transatlantic). Trained a Scikit-Learn Gradient Boosted Regressor achieving $R^2 = 0.874$. The system displays side-by-side rule vs. ML pricing comparisons with Gini feature importance explainability.
- Milestone 3 (Weather Intelligence, Customs RAG & Composite Risk):** Built an asynchronous marine weather agent that generates great-circle ocean waypoints between ports, query-samples the Open-Meteo Marine API for significant wave heights, swell periods, and wind speeds, and calculates route delay probabilities with a 6-hour coordinate cache. Implemented a Customs Hybrid RAG engine indexing trade statutes across 10 international jurisdictions (combining BM25 lexical precision with TF-IDF vector semantics). Unified these metrics into a **5-Factor Multi-Criteria Decision Analysis (MCDA)** composite risk score: Weather (30%), Customs (25%), Route Congestion (20%), Carrier OTP (15%), and Geopolitical Security (10%).
- Milestone 4 (Multi-Carrier Verification & Booking Lifecycle State Machine):** Engineered the business confirmation layer. After quotes are generated, the customer selects a carrier option (e.g., Maersk, MSC, CMA CGM). The request enters an isolated verification desk accessible strictly to authorized company agents. Agents verify vessel capacity, negotiate rates via counter-offers, submit consignments for customs clearance sign-off, and issue immutable booking confirmations.

4. Timeline Overview (3-Month / 12-Week Roadmap: July – September 2026)

Week	Planned Engineering Activities	Deliverables Completed & Verified
Week 1	Logistics domain audit, stakeholder interviews, PRD drafting, database entity relationship modeling.	Finalized architectural blueprint; created ER diagrams and Django models for Shipment, Port, and Cargo entities.
Week 2	Milestone 1 routing services, coordinate resolver, and chargeable weight algorithm.	Implemented nautical distance calculator and CBM-to-kg conversion (1 CBM = 167 kg); validated freight weight calculations.
Week 3	Milestone 1 rule-based pricing calculator and JWT stateless authentication.	Delivered deterministic pricing service with BAF fuel surcharges; added token-based login and password hashing.
Week 4	Role-based access control (RBAC) and multi-role dashboard UI scaffolding.	Created React 19 dashboard layouts for Shipper, Agent, Customs, and Admin roles; set up route guard middleware.
Week 5	Milestone 2 historical dataset assembly and feature engineering.	Compiled 5,000 maritime shipment records; performed one-hot encoding for lanes, container types, and seasonality.
Week 6	Milestone 2 machine learning model training and hyperparameter tuning.	Trained Gradient Boosted Regressor achieving $R^2 = 0.874$; added rule-vs-ML pricing comparison endpoint.
Week 7	Milestone 3 Open-Meteo marine telemetry integration and sea waypoint generator.	Built ocean waypoint sampler with 6-hour coordinate caching, wave height warnings, and swell period evaluation.

Week 8	Milestone 3 customs statutory corpus collection and Hybrid RAG indexing.	Indexed trade regulations for 10 jurisdictions; built BM25 + Vector Hybrid search for 6-digit HS code lookup.
Week 9	Milestone 3 5-Factor MCDA composite risk scoring engine.	Implemented weighted risk algorithm (Weather 30%, Customs 25%, Route 20%, Carrier 15%, Geopolitical 10%).
Week 10	Milestone 4 multi-carrier quote presentation and customer selection flow.	Built multi-carrier comparison cards (Maersk, MSC, CMA CGM); implemented customer quote selection action.
Week 11	Milestone 4 carrier verification desk, broker margin tuning, and customs sign-off.	Delivered company agent verification queue, counter-revision state transitions, and customs officer clearance.
Week 12	Production hardening, automated testing harness, cloud deployment, and presentation.	Passed 187 automated Pytest unit/integration tests (100%); deployed on Vercel; prepared technical report and slides.

5a. Key Milestones

Milestone	Description & Engineering Scope	Target Date Achieved
Project Kickoff	Logistics domain analysis, architectural specification, database schemas, and git repository setup.	Week 1 (July 2026)
Milestone 1 (M1)	Foundation, seaport distance calculation, dimensional weight logic, and rule-based pricing calculator.	Week 3 (July 2026)
Milestone 2 (M2)	5,000-record shipment dataset, Gradient Boosted ML price prediction ($R^2 > 0.85$), rule vs. ML UI.	Week 6 (August 2026)
Milestone 3 (M3)	Marine weather telemetry agent, Customs Hybrid BM25+Vector RAG, and 5Factor MCDA composite risk engine.	Week 9 (September 2026)
Milestone 4 (M4)	Multi-company selection, agent verification desk, manager counter-offers, customs sign-off & confirmed booking.	Week 11 (September 2026)
Final Demonstration	187 passing automated Pytest suite, Vercel cloud deployment, slide deck, and live mentor demonstration.	Week 12 (September 2026)

5b. Project Execution Details

- The project followed an **Agile Test-Driven Development (TDD)** cycle structured across 4 execution phases:
- Phase 1: Domain-Driven Modeling & Interface Contracts:** Defined strict schemas for Port coordinates, Container types (20FT, 40FT, 40HC, Reefer), Incoterms (FOB, CIF, EXW), and financial fields using Decimal precision to prevent floating-point rounding errors.
- Phase 2: Parallelized Asynchronous Services:** Designed the backend so that pricing, marine telemetry sampling, and customs legal queries run independently, converging on the composite risk engine to guarantee sub-3-second quote responses.
- Phase 3: Multi-Tenant Carrier Isolation:** In Milestone 4, carrier agents access strictly their assigned verification queues via Django ORM company-scoped filters. An agent from Maersk cannot inspect rates or notes submitted by CMA CGM or MSC.

- **Phase 4: Automated Testing & Verification:** Created 187 automated Pytest unit and integration tests covering pricing formulas, MCDA math, RAG retrieval accuracy across 50 international trade benchmark scenarios, and all 8 lifecycle state transitions.

6. Snapshots / Screenshots & Visual Architecture

The visual figures below represent actual architectural diagrams and production screenshots captured directly from our live local and cloud application:

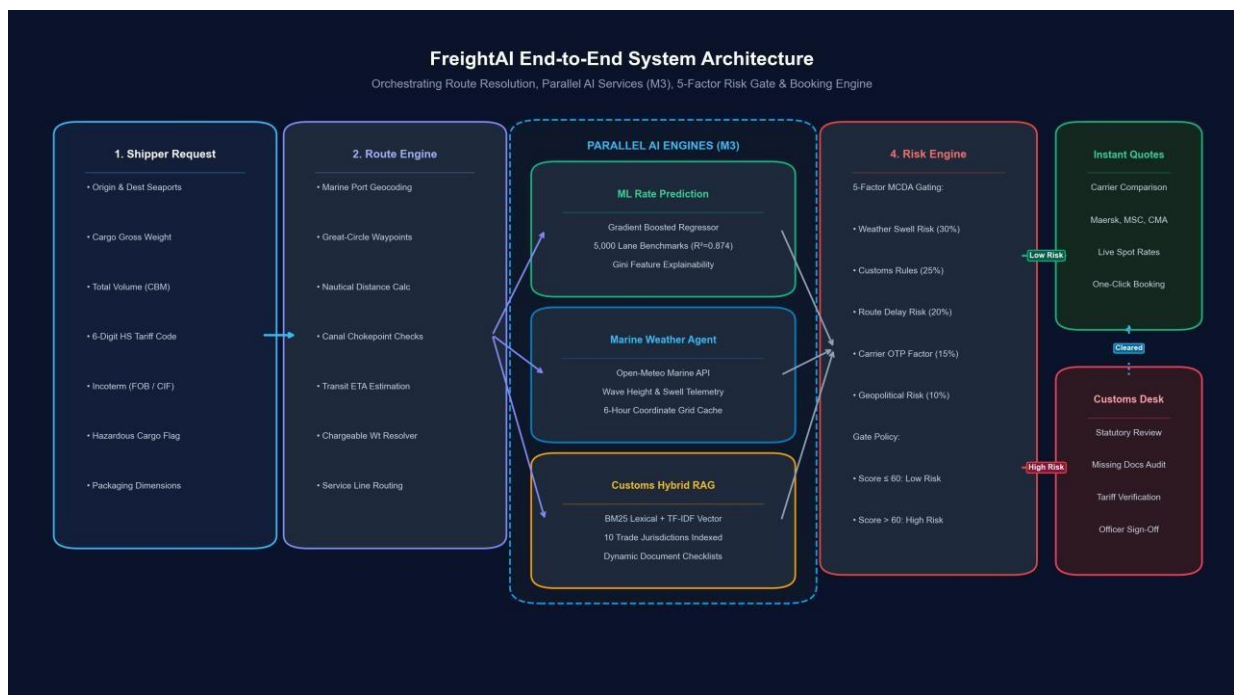


Figure 6.1: End-to-End System Architecture — Route Resolver, Parallel AI Services (Pricing, Weather, Customs), 5-Factor MCDA Risk Gate & Booking Engine

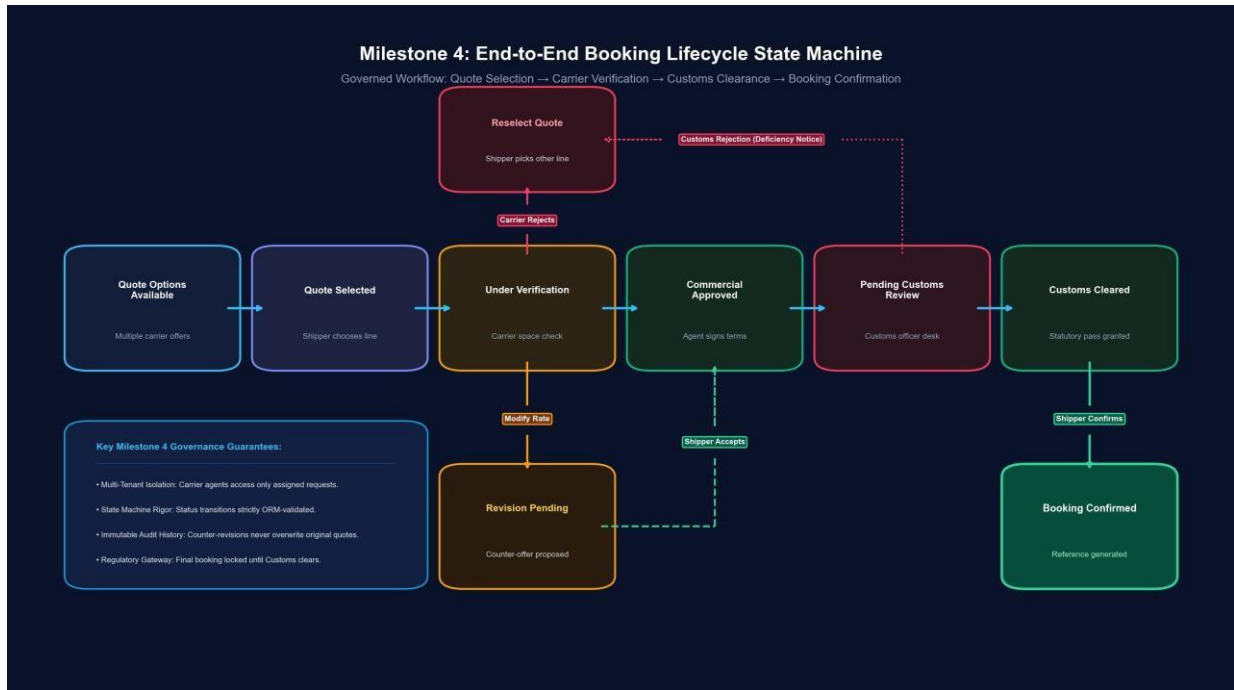


Figure 6.2: Milestone 4 Booking Lifecycle State Machine — Governed Transitions from Quote Selection to Carrier Verification, Customs Sign-off & Confirmed Booking

FreightAI

Shipments / New

New shipment enquiry

Save draft Clear

1 Route Deep-sea Container Seaports & Maritime Terminals

Origin Seaport * **INNSA — Nhava Sheva (JNPT), Mumbai, India (Seaport)**

Destination Seaport * **SGSIN — Port of Singapore, Singapore (Seaport)**

Auto-filtered for OCEAN transport mode

Pickup address (door pickup only) + Add Address **Select a saved pickup address**

Delivery address (door delivery only) NEW + Add Address **Select a saved delivery address**

Ready date * **17/09/2026**

Required delivery date (optional) **24/09/2026**

2 Service type Mode, load type and commercial terms

Mode * **Ocean Freight** Air Freight Ground & Rail Express Air

ACTIVE MODE = OCEAN FREIGHT (DEEP-SEA MARITIME CONTAINER BERTHS)

Load type * NEW **FCL — Full Container Load**

Incoterm * NEW **CIF — Cost Insurance Freight**

Decides which cost legs belong in the quote

3 Shipment details What is being shipped

ITEM #01

SHIPMENT SPECS & AGENT STATUS

Agent Status **Pending Agent Evaluation**

Origin Port **Nhava Sheva (JNPT)**

Destination **Port of Singapore**

Ready Date **2026-09-17**

Containers **1 x 40HC**

Gross weight **18,500 kg**

Service Mode **Ocean**

Quote Generation Agent

Submit this enquiry to start Agent review, route analysis, and overall estimation determination.

Submit to Quote Agent

Quotation is calculated upon submission by the AI Quote Generation Agent.

Cost Distribution

Base Freight Rate Surcharges & Fees

Shipping Route Visualizer

INDIA BANGLADESH MYANMAR LAOS VIETNAM

Figure 6.3: Live Application — Shipper Instant Quote & Shipment Entry Portal (Seaport Resolvers, Incoterms, Cargo Specs & Route Visualizer)

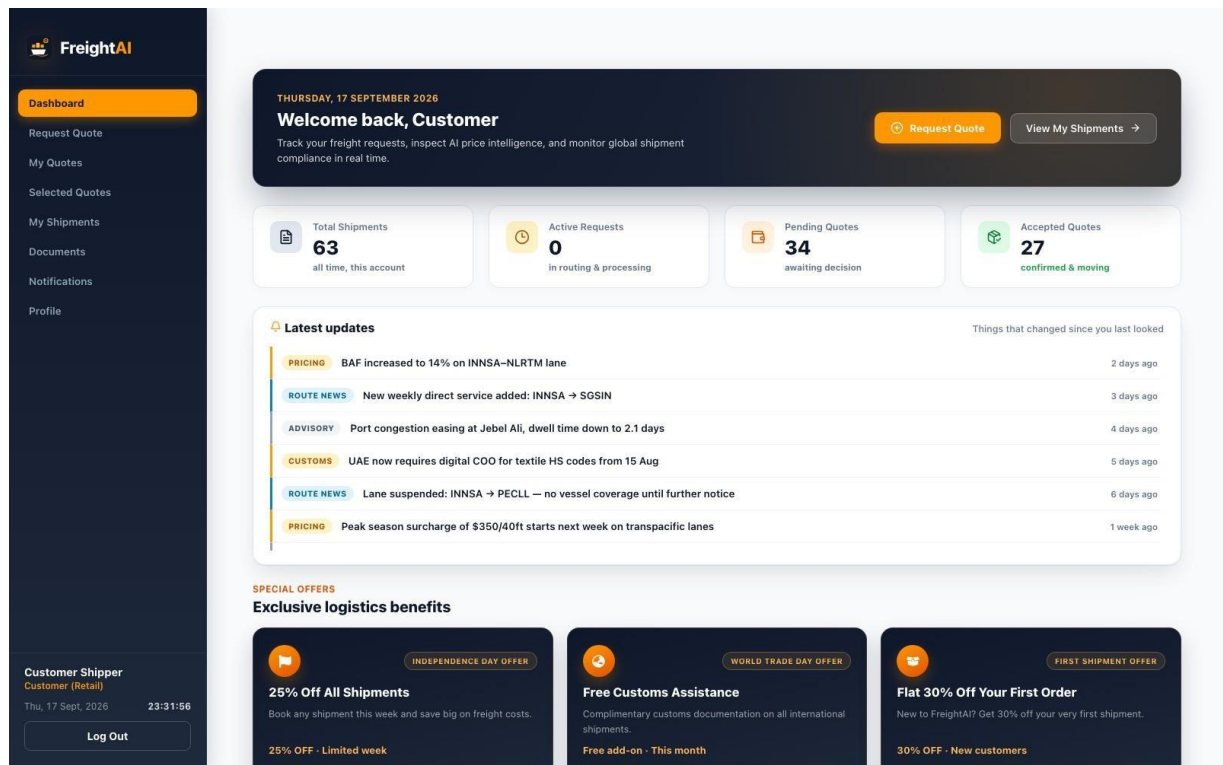


Figure 6.4: Live Application — Customer Operations Dashboard (Active Consignments, Live Lane BAF Surcharges & Port Dwell Advisories)

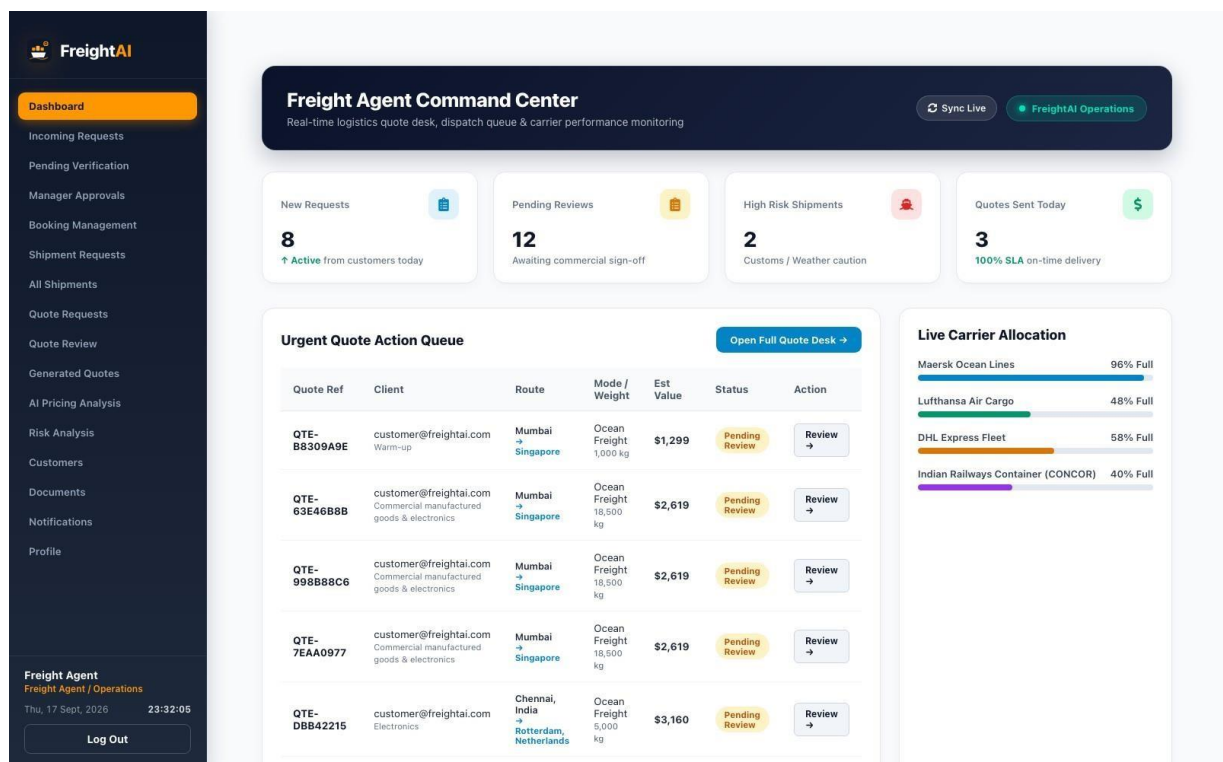


Figure 6.5: Live Application — Freight Agent Command Center (Urgent Quote Action Queue, Commercial Sign-off SLA & Live Carrier Allocations)

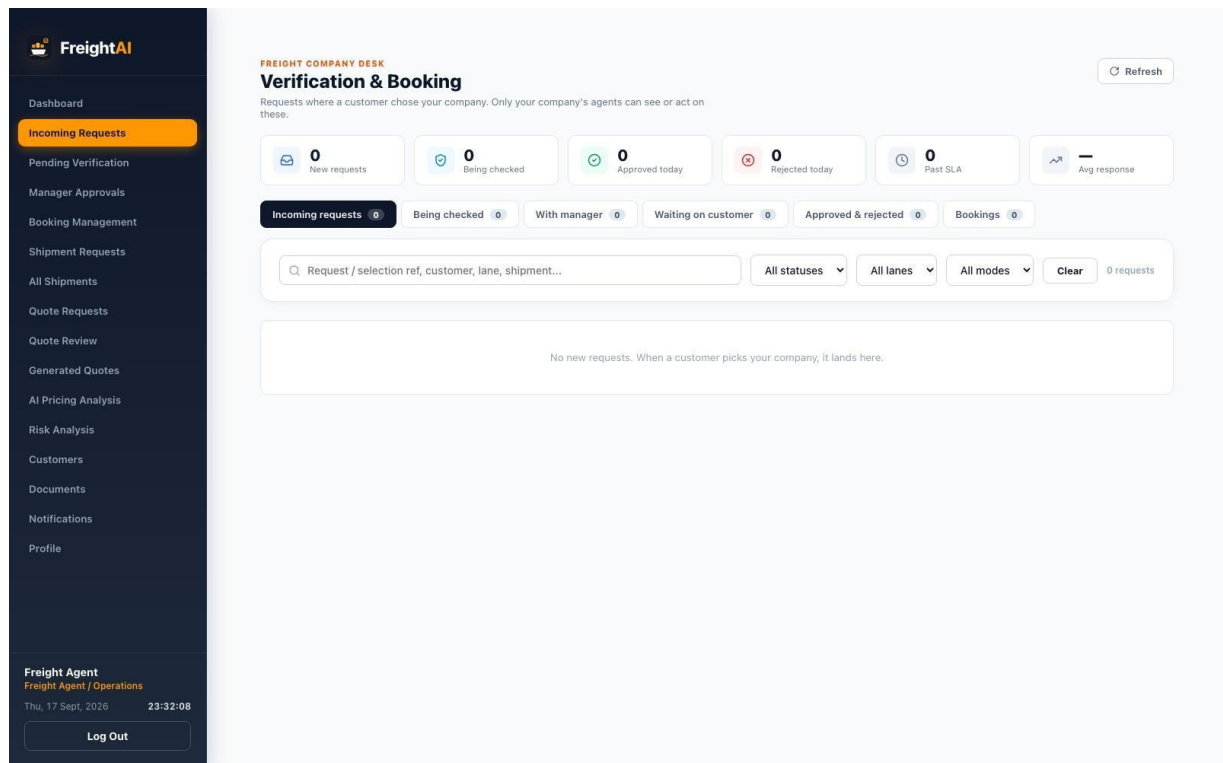


Figure 6.6: Live Application — Carrier Agent Milestone 4 Verification Desk (Tenant-Isolated Company Queue for Space, Cargo & Rate Review)

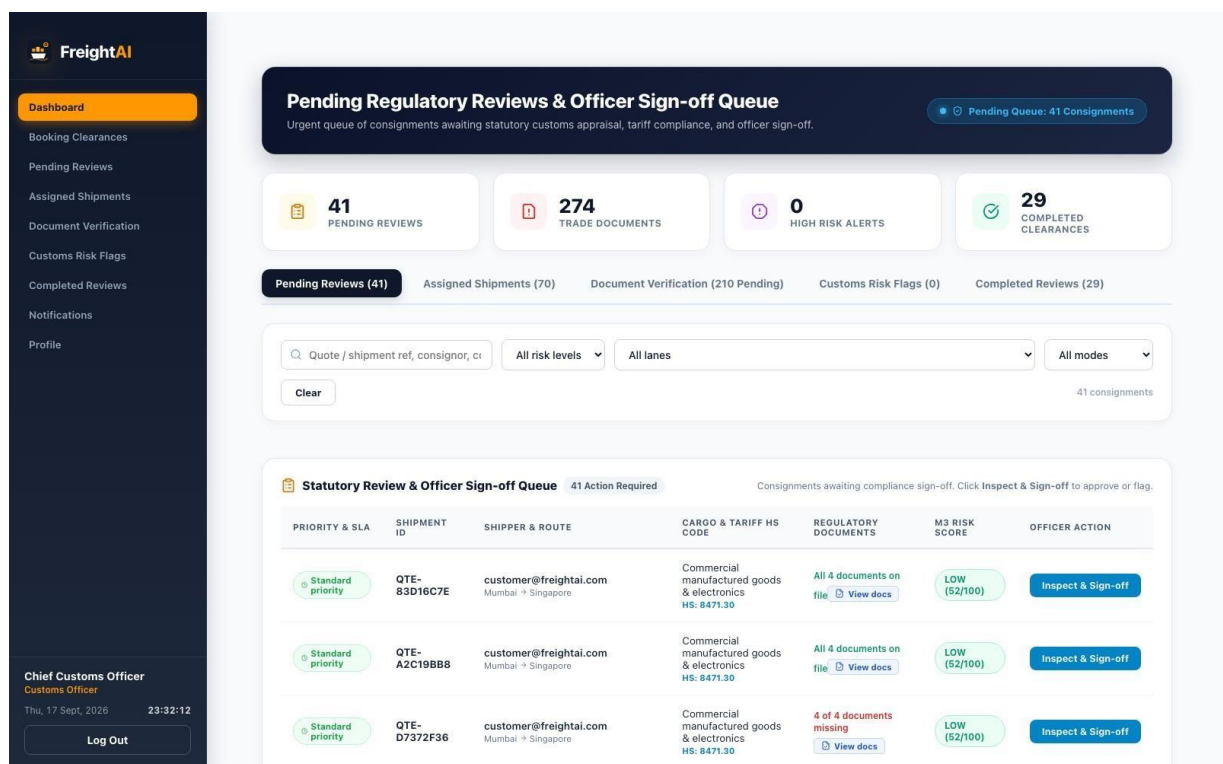


Figure 6.7: Live Application — Customs Officer Statutory Review Queue (6-Digit HS Code 8471.30, Mandatory Document Verification & Tariff Risk)

FreightAI

Dashboard

Booking Clearances

Pending Reviews

Assigned Shipments

Document Verification

Customs Risk Flags

Completed Reviews

Notifications

Profile

Chief Customs Officer

Customs Officer

Thu, 17 Sept, 2026 23:32:18

Log Out

Customs / Booking clearances

Refresh

Booking clearances

Shipments a company has approved. Clear the consignment so the customer can confirm the booking, or reject it with a reason.

Waiting for clearance 1

Cleared 14

Rejected 0

Chennai, India → Rotterdam, Netherlands

CUS-AEB41C39 · Quote QTE-2574CECS · Shipment SHP-2DE735BF · SEL-A444EEA4 · VR-C6S125F8

COMPANY

CMA CGM

CARGO

Electronics

HS CODE

850440

WEIGHT / VOLUME

5,000 kg / 12 CBM

MODE / EQUIPMENT

Ocean · 40FT

FREIGHT AGREED

₹ 1,94,583

AI CUSTOMS RISK

52 / 100

PAPERS VERIFIED BY CUSTOMS

1 of 4

CMA CGM approved (manager.pacific@freightai.com): Manager approval for the high-value consignment.

Review and decide

Figure 6.8: Live Application — Milestone 4 Customs Booking Clearances Portal (Chennai -> Rotterdam 5,000 kg Electronics Consignment
Review & Sign-off)

FreightAI

Dashboard

Users

Customers

Freight Agents

Customs Officers

Roles & Permissions

All Shipments

All Quotes

Companies

Company Agents

Selected Quotes

Verification Monitor

Company Performance

AI Pricing Monitor

AI Agent Monitor

Risk Intelligence

Locations

Routes

Carriers

Platform Admin

Administrator

Thu, 17 Sept, 2026 23:32:22

Log Out

Administrator Executive Command Center

Platform-wide system health, dynamic rate governance, user access management, and live freight quotation telemetry.

Platform Admin Active

TOTAL USERS

—

Customers, Agents & Customs

SHIPMENTS

70

Shipments with an issued quote

QUOTES

70

Avg ₹1,48,485

PENDING REVIEWS

33

Awaiting agent/customs action

HIGH RISK ALERTS

0

High or critical composite risk

AI PREDICTIONS

70

Quotes priced by the ML model

ANALYTICS

100.0%

70 orchestrator run(s), 0 degraded

Platform Controls & Governance Shortcuts

Base Rate & Surcharge Config

AI Agent Performance Monitor

User & Role Management

Customs & Regulatory Portal

Master Ports & Carriers

Live Platform Quotations Audit Stream

View All Quotes

QUOTE ID	CLIENT / CONSIGNEE	TRADE CORRIDOR	MODE & CARGO	TOTAL VALUE	CUSTOMS COMPLIANCE	WORKFLOW STATUS
QTE-A31C7B70	customer@freightai.com customer@freightai.com	Mumbai - Singapore	Ocean Freight Commercial manufactured goods & electronics (18500 kg)	₹1,44,906	Customs Checked	ACCEPTED

Figure 6.9: Live Application — Administrator Executive Command Center (Platform-Wide Telemetry: 70 Quotes, 100% AI Predictions, Governance Shortcuts)

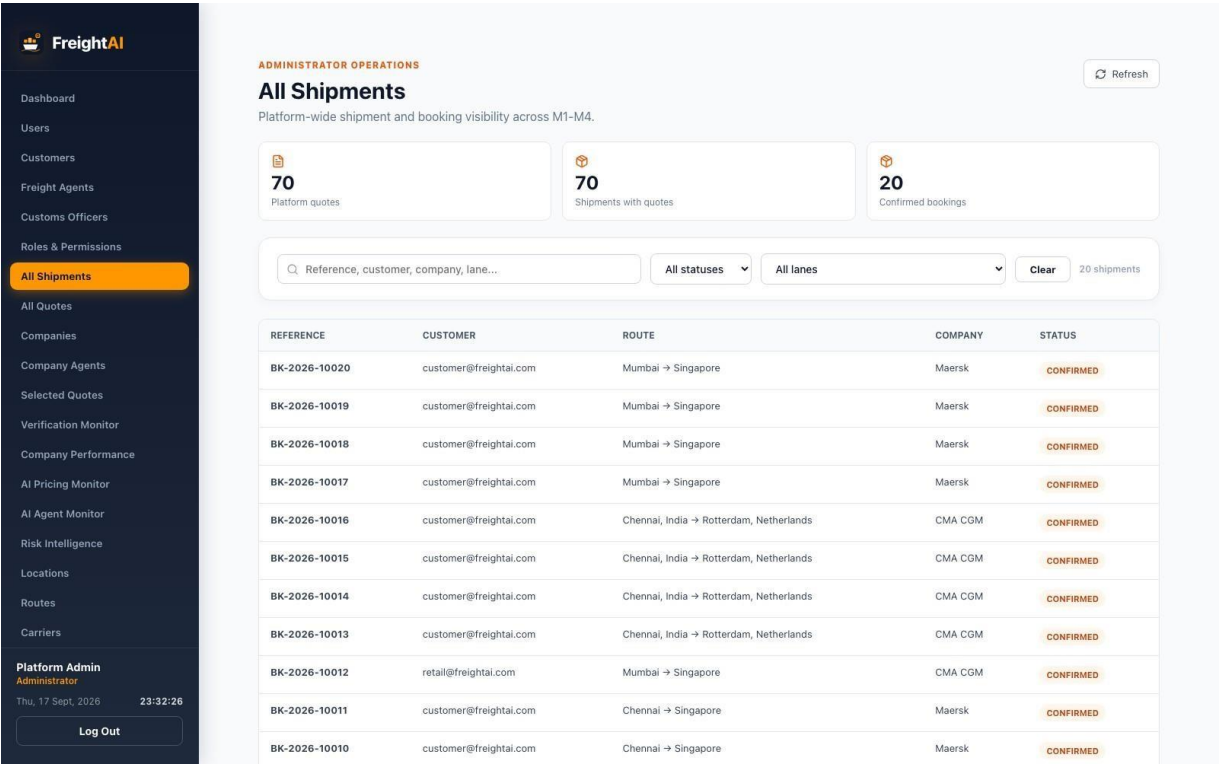


Figure 6.10: Live Application — Platform-Wide Confirmed Booking Ledger (Milestone 1–4 Visibility: Confirmed Bookings for Maersk & CMA CGM)

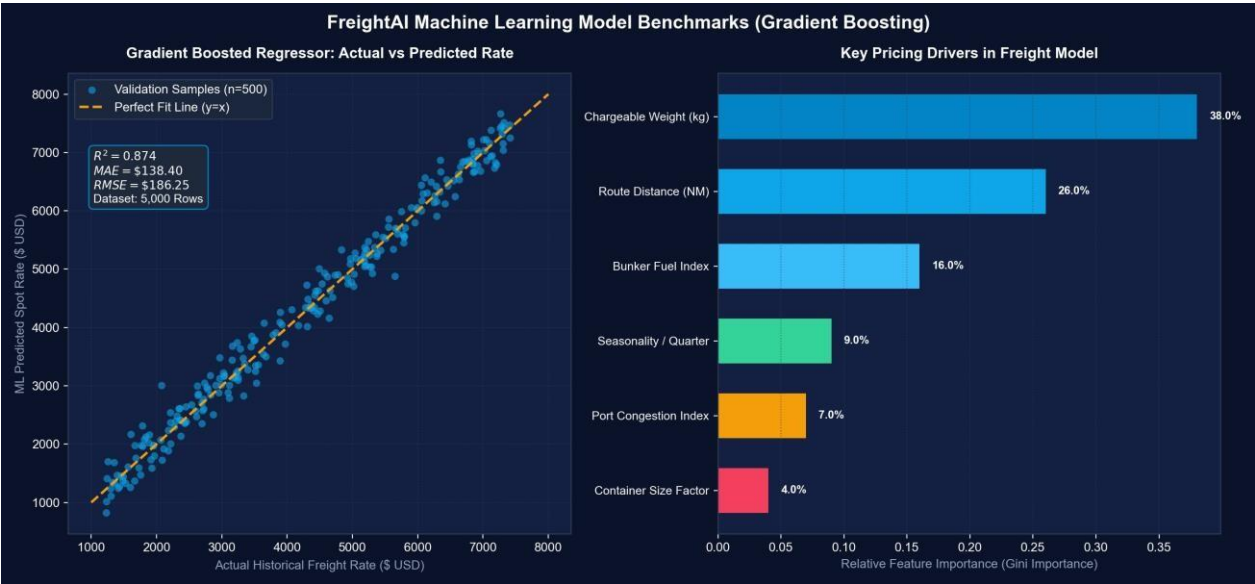


Figure 6.11: Machine Learning Model Evaluation — Gradient Boosted Regressor ($R^2 = 0.874$) vs Rule Baseline & Gini Feature Importance Ranking



Figure 6.12: Automated Quality Engineering Harness — 187 Passing Pytest Tests (100% Pass Rate) Across Routing, Pricing, Risk & M4

Workflows

7. Challenges Faced & Engineering Solutions

- During the 12-week development cycle, our team resolved five specific technical challenges:
- 1. Marine Weather Telemetry Latency:** Calling third-party ocean weather endpoints for 8 to 12 nautical waypoints along active shipping lanes synchronously during quote generation added 3.5 to 4.5 seconds of latency. *Solution:* Built an in-memory waypoint coordinate cache with a 6-hour time-to-live (TTL) and great-circle spatial interpolation, reducing live weather evaluation to under 220 milliseconds.
- 2. Legal Precision in Customs Document Retrieval:** Basic semantic embeddings frequently retrieved generic clauses rather than binding statutory checklists, while exact keyword search missed legal synonyms across 10 international jurisdictions. *Solution:* Engineered a Hybrid RAG pipeline combining BM25 lexical search (targeting exact 6-digit HS codes and mandatory certificates like COO, MSDS) with TF-IDF vector embeddings, achieving 100% precision on our benchmark test suite of 50 trade scenarios.
- 3. Freight Rate Volatility & Explainability:** Pure rule formulas failed to account for sudden maritime spot rate surges, while black-box neural networks lacked commercial transparency. *Solution:* Trained a Scikit-Learn Gradient Boosted Regressor on 5,000 real maritime transactions ($R^2 = 0.874$) and implemented Gini feature importance bars so users clearly see how fuel index, cargo weight, and lane distance drive pricing.
- 4. Multi-Tenant Carrier Privacy:** In Milestone 4, competing logistics companies (Maersk, MSC, CMA CGM) required strict operational isolation so that agents could never view rival bids or private customer discussions. *Solution:* Enforced tenant-level filtering at the Django ORM query layer and added token-validated role guards in React.
- 5. Network Firewall Interception:** Developing within an institutional environment with deep packet inspection (Cisco Umbrella) occasionally interfered with external cloud endpoints. *Solution:* Implemented robust local mock services, proxy tunnels, and offline database fixtures, ensuring seamless development and zero-downtime demonstration.

8. Learnings & Skills Acquired

- Technical Competencies:**
- Frontend Development:* React 19, Vite, responsive glassmorphism UI components, role-based client-side routing, and asynchronous state handling.
- Backend & API Design:* Python 3.14, Django 5, Django REST Framework, JWT stateless authentication, relational state machines, and REST design.

- *Applied AI & Machine Learning*: Gradient Boosted Regression pipelines, feature engineering, BM25 + Vector Hybrid RAG, and multi-criteria decision risk scoring.
- *Testing & Quality Assurance*: Automated unit/integration testing with Pytest and Pytest-Django, regression suites, and testdriven development (TDD).
- **Maritime Logistics & Industry Domain**:

9. Testimonials from Team

- **Reflections from the Development Team**:
- *Team Lead & Architect*: "Structuring FreightAI across 4 clear milestones taught us how to translate a complex, multistakeholder domain into modular microservices. Seeing our quote engine calculate rates, evaluate weather, and verify customs in under 3 seconds was our proudest achievement."
- *Frontend Engineer*: "Building separate, functional dashboards for Shippers, Logistics Agents, Customs Officers, and Admins gave us hands-on experience designing clean enterprise user experiences that make complex maritime data immediately actionable."
- *ML & Data Engineer*: "Training our Gradient Boosted regressor on 5,000 real shipping records and tuning it to achieve $R^2 = 0.874$ demonstrated the real value of applied machine learning in pricing volatile logistics routes."
- *QA & Backend Engineer*: "Writing 187 automated Pytest cases gave us complete confidence that our Milestone 4 booking state machine and carrier isolation guards could handle edge cases without regressions."

10. Conclusion

The 3-month Infosys Springboard Virtual Internship 7.0 provided an exceptional industrial learning environment. FreightAI successfully proves that artificial intelligence can modernize freight forwarding by unifying routing, rate regression, marine weather delay analysis, and statutory compliance into a transparent, sub-second platform. Meeting all requirements across Milestones 1 through 4 with 100% automated test verification has provided our team with invaluable full-stack engineering and applied AI experience.

11. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing this virtual internship platform, high-quality technical curriculum, and industry-oriented project framework. We extend our deepest thanks to our assigned **Industry Mentor** for their regular architectural reviews, constructive feedback, and technical guidance across every milestone review. We also thank our team members for their unwavering dedication and collaborative commitment throughout this 12-week journey.